Zachary Dietderich

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Education (Graduating May 2026)

Bachelor of Science in Mechanical Engineering with Mathematics minor from San Diego State University GPA: 3.64

Automotive Engineering courses and shopwork (August 2019 – May 2022)

- Completed automotive engineering coursework with hands-on shop experience in vehicle systems, including engines, brakes, and suspension.
- Applied diagnostic, maintenance, and repair techniques using industry-standard tools while following proper safety and shop procedures.

Work Experience

Mechanical Engineering Intern - Lawrence Berkeley National Laboratory May 2025 - August 2025

- Evaluated new impregnation materials for superconducting magnets using single-fiber pull (SFP) and 3-point bending tests to characterize mechanical and thermal performance.
- Designed and 3D-printed TPU (95A HF) molds for curing resin/wax samples; conducted SFP testing at room temperature and 4.3 K (liquid helium).
- Improved an automated superconducting magnet winder by converting legacy Excel-based calculations to Python, increasing computational efficiency.
- Developed **Python** scripts to validate automated winding parameters and designed/printed **Creo**-modeled mandrel prototypes for dipole, quadrupole, and combined function coils.
- Led a technical seminar presenting project design, methods, and results to scientists and engineering teams.

Student Engineering Intern - Applied Spectra Inc. May 2024 - August 2024

- Conducted energy stability testing of Pulsed Nd:YAG laser spectroscopy systems to optimize performance.
- Collected and analyzed energy data under varying ambient temperatures to determine system efficacy under varying conditions.
- Gained hands-on experience in web development using **C#**, contributing to proprietary licensing software for global instrumentation.

Mechanical Engineering Intern - Lawrence Berkeley National Laboratory May 2023 - August 2023

- Designed a test fixture and implemented materials testing to analyze the permanent deformation (creep) at high temperatures in inert environments (argon gas).
- Developed a **LabVIEW** program to display real-time temperature vs. time and log the temperature data during the vacuum impregnation cycle for coils to be used in superconducting magnets.
- Assisted with winding and assembly of the superconducting coils into a magnet to be used for proton and carbon radiation therapy to treat non-operable cancers.

Projects (More projects shown in portfolio linked at top)

Aztec Baja Racing – Engine and Transmission Dynamometer (August 2025 – Present)

- Designed and built a small-engine dynamometer (≤15 HP) to measure engine and transmission performance by calculating horsepower from torque and RPM using a hydraulic pump load system.
- Developed a custom **DAQ** system using **Arduino**, integrating a **Hall effect sensor** with a magnet-based 3D-printed shaft assembly for RPM measurement, along with **load cell,** for engine torque, and **pressure transducer** sensors.
- Utilized the dynamometer to evaluate CVT efficiency by replacing a legacy CVT setup, enabling comparative **analysis of power transmission and for an increase in system performance**.

ASML – Student Affiliate (August 2025 – Present)

- Actuating stage and camera system for vacuum chamber testing hydrogen radical and tin reactions modeled and assembled using **SolidWorks**.
- Used mechanical analysis to determine validity of prototyping and engineering design through **FEA analysis**.

Technical Skills

Engineering & CAD: SolidWorks, Creo Parametric, Onshape, Finite Element Analysis (FEA), Ansys Mechanical
Programming & DAQ: MATLAB, Python, LabVIEW, Arduino, C#, Excel
Manufacturing & Materials: Powder Metallurgy, 3D Printing, MIG Welding, Epoxies, Material Characterization, GD&T